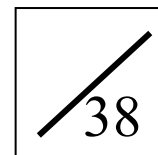


Pre S3 L03 Inequality Quiz
(ANSWER)

Name: _____

Date: _____



1. Find the sum of all the negative integers x satisfying the inequality $x > -2\pi$.

A. -21π

B. -21

C. -5

D. 0

The answer is B.

$$x > -2\pi$$

$$x > -6.28\ldots$$

$\therefore$ The negative integers satisfying the inequality are $-6, -5, -4, -3, -2$, and -1 .

$\therefore$ The required sum $= (-6) + (-5) + (-4) + (-3) + (-2) + (-1)$

$$= -21$$

B

2. If $2x \leq 3$, which of the following is not true?

A. $4x \leq 2x + 3$

B. $1 - 2x \leq -2$

C. $2x + 3 \leq 6$

D. $\frac{2x}{3} \leq 1$

The answer is B.

A: $2x \leq 3$

$$2x + 2x \leq 2x + 3$$

$$4x \leq 2x + 3$$

B: $2x \leq 3$

$$-2x \geq -3$$

$$1 - 2x \geq 1 - 3$$

$$1 - 2x \geq -2$$

$\therefore$ The given inequality is not true.

C: $2x \leq 3$

$$2x + 3 \leq 3 + 3$$

$$2x + 3 \leq 6$$

D: $2x \leq 3$

$$\frac{2x}{3} \leq \frac{3}{3}$$

$$\frac{2x}{3} \leq 1$$

$\therefore$ B is not true.

B

3. If $x \leq y$ and $z > y$, where z is a negative number, which of the following is not true?

A. $x \leq y < z$

B. $y - x < z - x$

C. $xz < xy$

D. $-\frac{x}{4} < -\frac{z}{4}$

The answer is D.

A: $\because x \leq y$ and $y < z$.

$$\therefore x \leq y < z$$

B: $\because y < z$

$$\therefore y - x < z - x$$

C: $\because x \leq y < z$ and $z < 0$.

$$\therefore x < 0$$

$$\therefore y < z \text{ and } x < 0.$$

$$\therefore xz < xy$$

D: $\because x \leq y < z$

$$\therefore -x > -z$$

$$-\frac{x}{4} > -\frac{z}{4}$$

$\therefore$ The given inequality is not true.

$\therefore$ D is not true.

D

4. Solve the inequality $3(x + 2) < 5(6 - x)$.

A. $x > 3$

B. $x < 3$

C. $x > -3$

D. $x < -3$

The answer is B.

$$3(x + 2) < 5(6 - x)$$

$$3x + 6 < 30 - 5x$$

$$3x + 5x < 30 - 6$$

$$8x < 24$$

$$x < 3$$

B

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5. Which of the following groups of numbers can all satisfy the inequality $7x + 3 > 17$?

- A. $-1, 0, 1$
- B. $1, 3, 5$
- C. $2, 3, 4$
- D. $2.1, 3.2, 4.3$

The answer is D.

R.H.S. = 17

When $x = 2.1$,

L.H.S. = $7(2.1) + 3 = 17.7 > \text{R.H.S.}$

When $x = 3.2$,

L.H.S. = $7(3.2) + 3 = 25.4 > \text{R.H.S.}$

When $x = 4.3$,

L.H.S. = $7(4.3) + 3 = 33.1 > \text{R.H.S.}$

$\therefore 2.1, 3.2$ and 4.3 can all satisfy the inequality.

D

6. (a) Solve the inequality $\frac{x}{4} - 2 \leq 0.8 - \frac{x}{3}$.

(3 marks)

(b) Write down all positive integers satisfying the inequality in (a).

(2 marks)

(a) $\frac{x}{4} - 2 \leq 0.8 - \frac{x}{3}$

$$12\left(\frac{x}{4} - 2\right) \leq 12\left(0.8 - \frac{x}{3}\right)$$

1M

$$3x - 24 \leq 9.6 - 4x$$

$$3x + 4x \leq 9.6 + 24$$

1M

$$7x \leq 33.6$$

$$x \leq \frac{33.6}{7}$$

$$\underline{x \leq 4.8}$$

1A

(b) The required positive integers are 1, 2, 3 and 4.

1A+1A

7. Write down all positive integers satisfying the inequality $\frac{4x+3}{5} < \frac{1-3x}{6} + 4$.

(5 marks)

$$\frac{4x+3}{5} < \frac{1-3x}{6} + 4$$

$$30\left(\frac{4x+3}{5}\right) < 30\left(\frac{1-3x}{6} + 4\right)$$

1M

$$24x + 18 < 5 - 15x + 120$$

$$24x + 15x < 5 + 120 - 18$$

1M

$$39x < 107$$

$$x < 2\frac{29}{39}$$

1A

$\therefore$ The required positive integers are 1 and 2.

1A+1A

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- *8. A **non-negative** number x satisfies the equation $\frac{2x-k}{3} = \frac{5x-7}{4} + \frac{x+k}{6}$, where k is a positive integral constant. How many possible values of k are there? (8 marks)

$\frac{2x-k}{3} = \frac{5x-7}{4} + \frac{x+k}{6}$	
$12\left(\frac{2x-k}{3}\right) = 12\left(\frac{5x-7}{4} + \frac{x+k}{6}\right)$	1M
$8x - 4k = 15x - 21 + 2x + 2k$	
$8x - 15x - 2x = -21 + 2k + 4k$	
$-9x = -21 + 6k$	
$x = \frac{-21+6k}{-9}$	
$= \frac{7-2k}{3}$	1A
$\therefore x \text{ is non-negative.}$	
$\therefore \frac{7-2k}{3} \geq 0$	1M
$7 - 2k \geq 0$	1M
$-2k \geq -7$	1M
$k \leq 3\frac{1}{2}$	1A
$\therefore k \text{ is a positive integer.}$	
$\therefore \text{ The possible values of } k \text{ are 1, 2 and 3.}$	1A
<u>i.e. There are 3 possible values of k.</u>	1A